

Title: Project Registration & Progress Review**FF No. 180**

Department: Computer Engineering	Academic Year: 2025-26
Semester: SEM-IV	Group No: SY CS E - 12
Project Title:- ExpLung-U: Explainable Lung Abnormality Detection and Severity Analysis Using Deep Learning	
Project Area: Medical Image Analysis using Deep Learning and Explainable Artificial Intelligence (Health Care)	

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Project approved / Not approved

Guide

Project Coordinator

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PROJECT SYNOPSIS

INTRODUCTION:

Lungs disease is a broad term for many diseases that affect the functioning of the lungs including pneumonia, tuberculosis, COVID-19, and lung cancer that collectively impact millions across the globe. Timely and reliable identification of the abnormalities is particularly crucial in the interests of proper treatment and recovery. Traditionally, radiologists have hand-analyzed X-ray and CT images to detect whether a patient has lung infections or nodules. It is subjective, time consuming, and could differ from an expert to another.

Recent developments in the field of Artificial Intelligence (AI) and Deep Learning have enabled automated and highly accurate analysis of medical images. These are now coming down in price as a consumer commodity but the challenge is that they are all “black box:” systems that don’t really explain how they know what they know. This opaqueness causes doctors to lose trust.

In this paper, to address the above challenges, we present ExpLung-U, an explainable lung abnormality identification approach which integrates the operations of segmentation, classification, severity analysis and visualization. This system is able to detect diseases, to show where the abnormality is found and to indicate its severity. This enables doctors to make faster, safer and better informed decisions.

PROBLEM STATEMENT:

" To Develop an explainable AI-driven system for automated lung segmentation, abnormality detection, disease severity quantification, and clinical insights from X-ray and CT images.”

OBJECTIVES:

The key objectives of our project are as follows :

1. Design an automated lung segmentation model based on U-Net
2. Detection of abnormalities with CNN classifiers. Make explainability available through Grad-CAM heatmaps.
3. View lungs in 2D /3D mode
4. Create a simple web UI (user interface) to interact with the model.

LITERATURE PAPERS:

Sr No	Paper	Year of Publication	Problem Solved	Algorithm / Methods / Process	Results	Future Scope	Limitations
1.	Automated Chest CT Image Segmentation of COVID-19 Lung Infection based on 3D U-Net	2020	Automated lung & infection segmentation	3D U-Net, patch-based training	Dice: Lung 0.956, Infection 0.761	Severity scoring	Limited dataset, Scalability Reduction.
2.	Lung Tumor Segmentation in MRI Mice Scans using 3D nnU-Net	2024	Accurate tumor segmentation	nnU-Net, volumetric learning	3D > 2D performance	Apply to human CT	High compute cost, Not Designed for X ray.
3.	Advanced U-Net Architectures with CNN Backbones	2025	Improve lung cancer detection	U-Net + CNN backbones	Dice $\approx$ 0.95	Real-time deployment	Low explainataion, Operates Black Box Model.
4.	Tumor-Anchored Deep Feature Random Forests	2025	OOD detection	Deep features + Random Forest	AUROC > 93%	Safer clinical AI	Extra module, No Lung Segmentation.
5.	Layer-wise Anatomical Attention for Radiology Reports	2025	Improve anatomical grounding	Attention + segmentation masks	Higher coherence	Apply to CT	X-ray only, Overlapping.
6.	Med SAM-Based Lung Masking for CXR Classification	2025	Reduce background bias	Med SAM + CNN	Better interpretability	Adaptive masking	Context loss, No severity , No Clinical Insights.

7.	CT-LungNet: A Deep Learning Framework for Precise Lung Tissue Segmentation in 3D Thoracic CT Scans	2024	Precise lung tissue segmentation in 3D CT scans with reduced computation	2.5D slice representation, U-Net with InceptionV3 encoder, transfer learning, low-parameter design	Dice score up to 99.7% (LUNA16), strong generalization on unseen datasets	Real-time clinical deployment, integration with disease detection pipelines	FF No. 180 High dependency on CT slice consistency, limited explainability
8.	Automatic Lung Segmentation in Chest X-ray Images Using Improved U-Net	2022	Robust lung segmentation in complex chest X-ray images	Improved U-Net, EfficientNet-B4 encoder, residual blocks, LeakyReLU activation, Dice loss	~97–98% Dice coefficient, improved robustness on severe disease cases	Extension to multi-organ X-ray segmentation, lightweight mobile deployment	Limited to X-ray modality, reduced performance with heavy occlusion
9.	nnU-Net: Self-adapting Framework for Medical Image Segmentation	2021	Automatic configuration of segmentation pipelines for medical images	nnU-Net framework with dynamic architecture selection, preprocessing, and training strategy	State-of-the-art Dice scores across multiple medical datasets	Can be extended to multi-organ and multi-modal segmentation	High computational requirements, limited interpretability
10.	Attention U-Net: Learning Where to Look for the Pancreas	2018	Improve segmentation accuracy by focusing on relevant regions	U-Net with attention gates to suppress irrelevant background features	Improved Dice score over standard U-Net, better boundary localization	Adaptable to lung segmentation and abnormality-focused tasks	Increased model complexity, slower training
11.	Multi-task Learning for Lung Disease Detection and Segmentation	2020	Joint learning of lung segmentation and disease classification	Shared CNN backbone with segmentation and classification heads	Improved diagnostic accuracy and robust segmentation	Integration with severity scoring and explainable AI	Task interference risk, complex training process

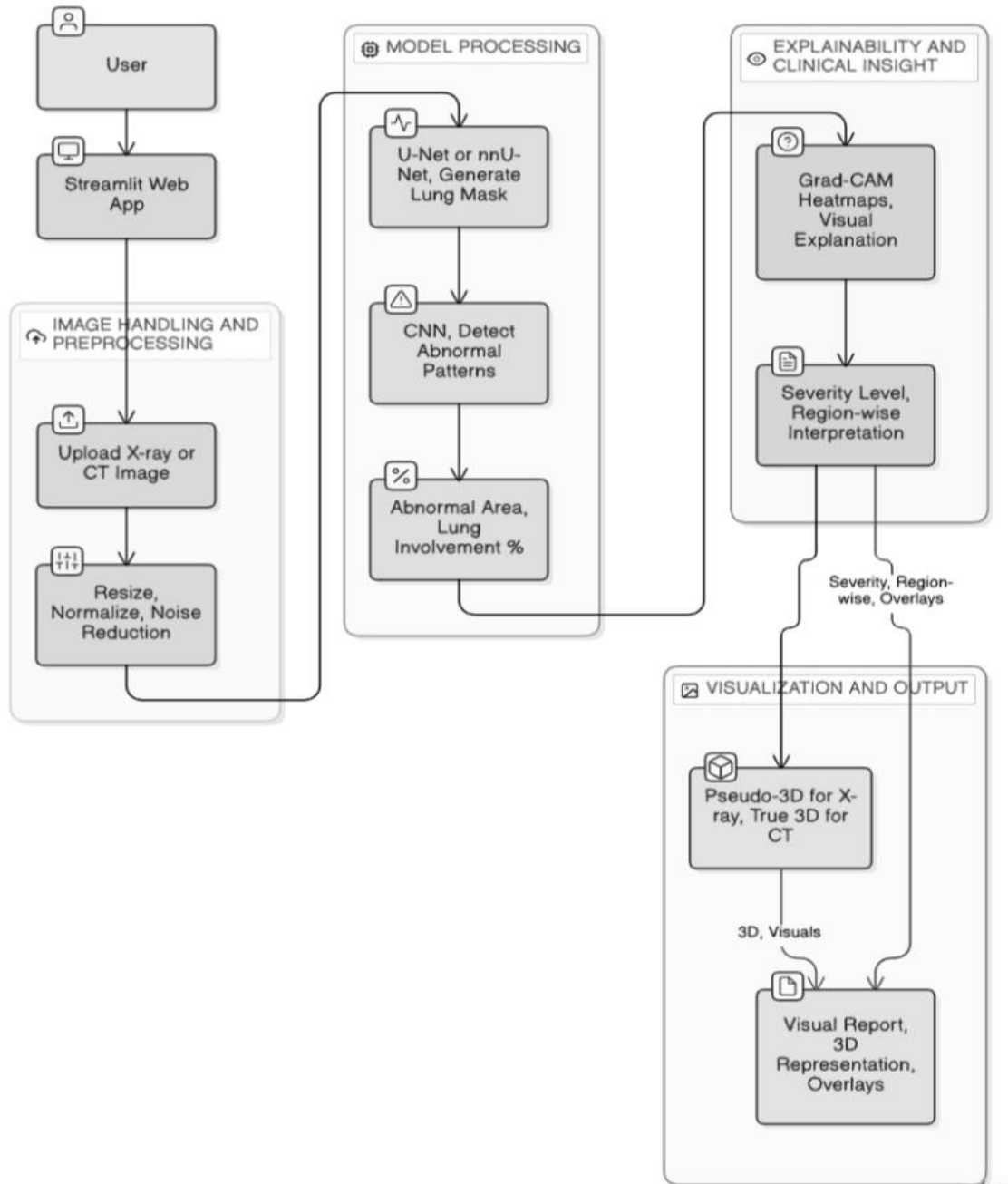
BLOCK DIAGRAM & FLOW CHART:

Fig.1.flow chart

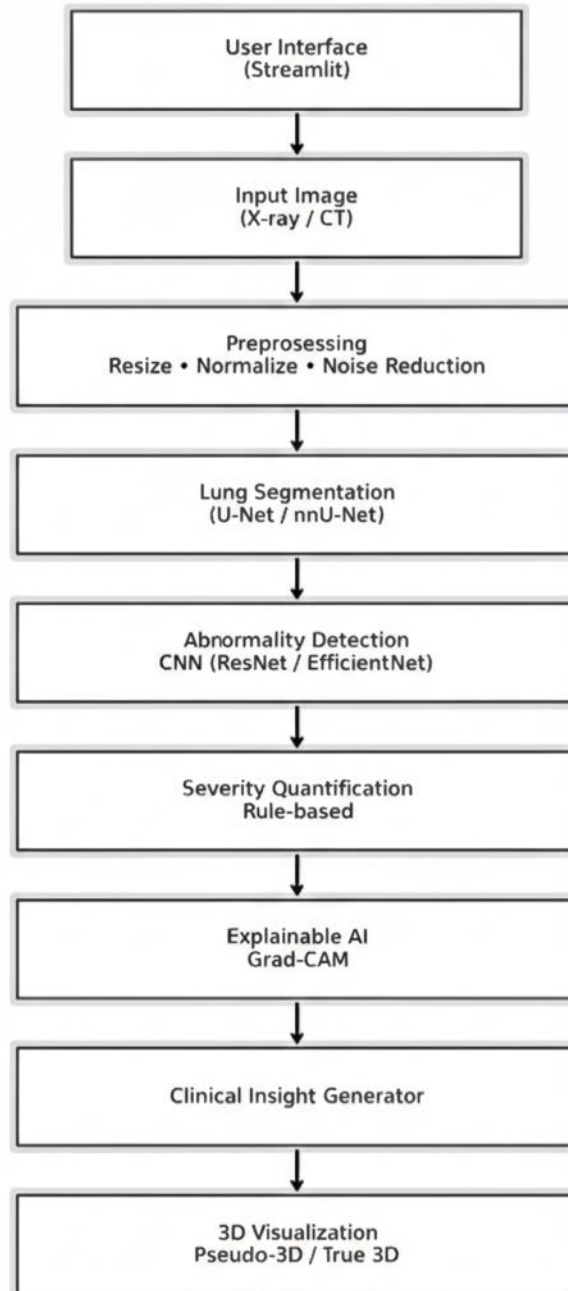


Fig.2.Block Diagram

METHODOLOGY:

1. Data Collection and Preprocessing

We source chest X-ray and CT images from conventional medical datasets. Preprocessing consists of image resizing, normalizing, and denoising, to enhance the image quality. We will be using NIH chest X-ray Dataset.

2. Lung Segmentation

A U-Net / nnU-Net model is trained to segment the lung regions from the input scans. The step eliminates non-lung background and enhances the concentration on the lung anatomy.

3. Lung Abnormality Detection

The segmented lung images are then processed by a CNN-based classifier like ResNet or EfficientNet. By learning discriminative lung features, the model identifies abnormality existence.

4. Severity Quantification

The lung abnormality is quantified by the area of the lung that is affected. This yields a numerical severity score that captures disease extent.

5. Explainable AI and Visualization

The Grad-CAM is then employed to visualize the influential regions for the model predictions. 2D and optional 3D visualization enhancement enables improved clinical interpretation.

6. User Interface and Output

A web interface enables users to upload images and get results via a browser. The system presents detection results, magnitude scores, and visual explanations.

Components:

- **Python** – Primary programming language for system design and integration.
- **OpenCV & NumPy** – Applied for image preprocessing, normalization and arithmetic operations.

- **PyTorch** – Used for image processing, normalization and operator overloading.
- **U-Net / nnU-Net** – Utilized to perform an accurate lung segmentation from X-ray/CT images.
- **ResNet / EfficientNet** – SampleCNN and SerieCNN used for lung abnormality.
- **Grad-CAM (Captum/TorchCAM)** – It delivers explainable AI by marking key areas of the images.
- **Plotly / PyVista** – Applied to 2D and 3D visualization of the lung anatomy and function.
- **Streamlit** – Web-based user interface for output presentation and user input.
- **SimpleITK (CT)** - The software is designed to import and read medical CT scans in either DICOM or NifTI file formats with true 3D analysis through preservation of slice order, spacing, and orientation.

Conclusion:

The ExpLung-U system shows that deep learning, explainable AI, can help doctors identify lung anomalies with speed and accuracy. The model offers predictions as well as explanations by integrating segmentation, classification, severity quantification, and visualization. It enhances trust and clinical decisions. The platform closes the divide between AI technology and practice in real-world medicine, accelerating and securing the diagnosis of lung diseases.

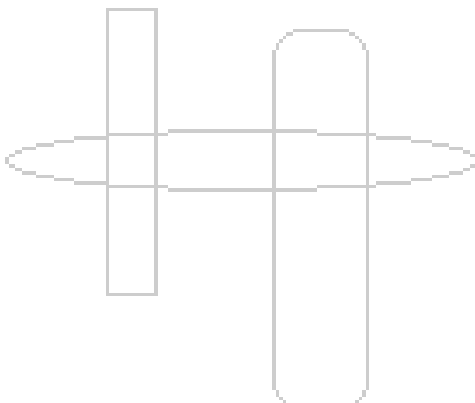
References:

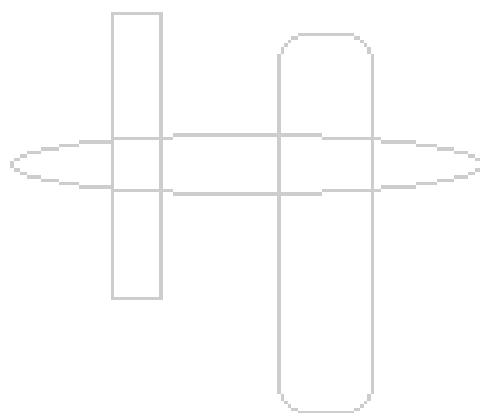
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Group No.			
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	Group No.:	Date:
Progress Review Report		
		
Signature of Guide:		

Review No.: 2**Group No.:****Date:****Progress Review Report****Signature of Guide:**